

**Studying Environmental
Conflicts with ParlaMint-TR
Parliamentary Corpus:
Environmental Discourse
Analysis of Turkish
Parliament(TGNA)**

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Theoretical Background

Environmental issues(such as wildfires, anti-HEPP movements, gold mining, and climate agreements) are frequently debated in the Turkish Grand National Assembly(TGNA).

Parliamentary speeches reveal how environmental crises are politically governed through chosen discourse.

These debates can be analyzed in terms of political consent-building and discursive framing.

Research question

1. How are environmental debates constructed in the Turkish Parliament?
 - 1.a. How environment environment-related debates in TGNA changed through time?
 - 1.b. Which parties are more involved in environmental issues in the parliament?
 - 1.c. Whether the frequency of these debates coincide with the environmental catastrophes in Turkey?
 - 1.d. How much time political parties dedicate to environmental issues compared to their total speaking time?

Hypotheses

- H1: JDP is more likely to frame environmental issues through development, innovation, infrastructure, and welfare than opposition parties.
- H2: Environmental justice-oriented discourse in parliamentary debates increases after grassroots/social mobilizations, but eventually declines.

Data: ParlaMint-TR Parliamentary Corpus & EJAtlas

“The current version of the corpus contains transcripts of debates in the last four terms (from 24 to 27) of the Grand National Assembly of Turkey, which is approximately 50 million words recorded in 1341 sessions from June 2011 to December 2022.”

- Web Scraping from TGNA website, transcripts HTML files
- Tokenization, sentence segmentation, morphological tagging and disambiguation were performed using TRmorph
- All available on GitHub

EJAtlas-Protest data to identify the times of protests and linked them to shifts in speeches

EJAtlas

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Global Atlas of Environmental Justice

Turkey

MapTiler Basic, OpenStreetMap, CC, ejolt, ENVIRONMENTAL JUSTICE, icta, erc, Syna, maptiler, Homs, Tripoli

EJAtlas - Global Atlas of Environmental Justice

38 cases have been filtered

The EJAtlas is a work in progress. Newly documented cases and information are continuously added to the platform. However, many are still undocumented and new ones arise. Please note that the absence of data does not indicate the absence of conflict.

Filters

Mobilizi... = Street prot...

and Country = Turkey

+

APPLY

CLEAR

> Legend

> Browse Maps

> Newly published featured maps

> Recently updated conflicts

Kanal Istanbul, Turkey

Methodology:

Dictionary Methods

- Filtering the data to identify environment-related speeches
- Keywords related to energy, extractivism, ecology, environmental justice
- Aggregating the data by year, party name and political orientation.

```
ENV_KEYWORDS = {
    "extractivism": [
        "madencilik", "maden", "altın madeni", "taş ocağı",
        "kömür", "linyit", "rezerv", "ruhsat", "altın"
    ],

    "energy_projects": [
        "hes", "hidroelektrik", "termik santral",
        "nükleer santral", "enerji yatırımı",
    ],

    "developmentalist": [
        "kalkınma", "büyüme", "yatırım",
        "istihdam", "milli ekonomi", "milletimizin refahı",
        "dışa bağımlılık"
    ],

    "sustainability": [
        "çevre", "doğa", "ekosistem",
        "biyoçeşitlilik", "iklim değişikliği",
        "sürdürülebilirlik", "politika yapıcılar"
    ],

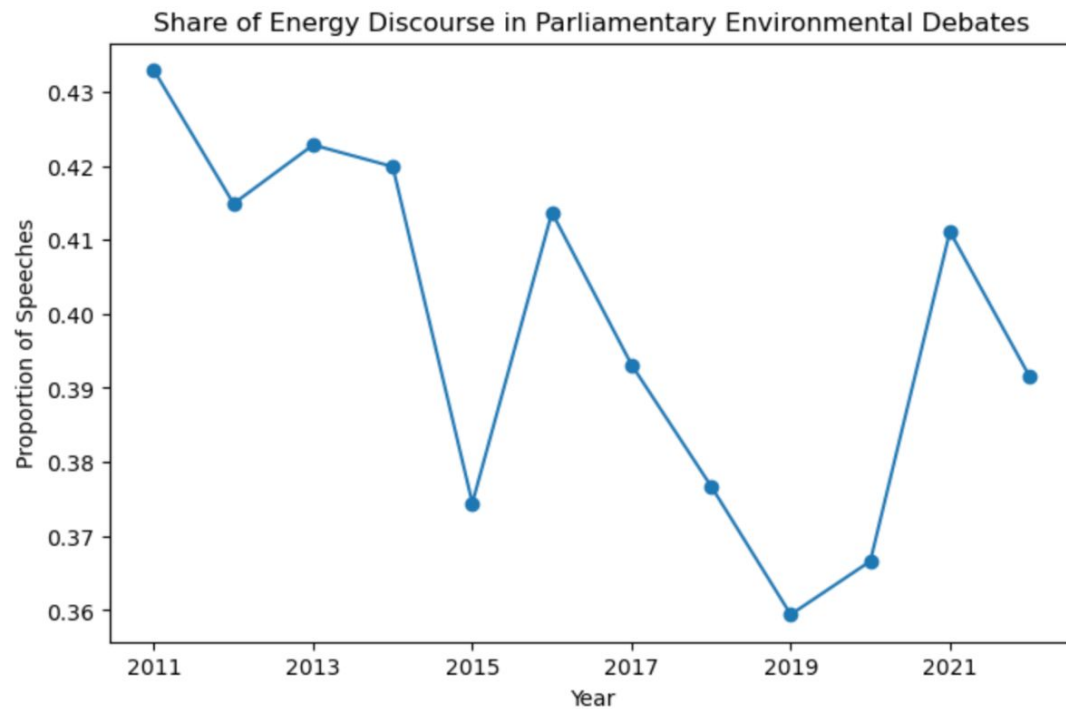
    "environmental_justice": [
        "çevre adaleti", "yaşam alanı",
        "yerel halk", "hak mücadelesi", "direniş",
    ]
}
```


Results

[46]: categories developmentalism energy environmental_justice extractivism sustainability

year					
2011	882	632	25	969	125
2012	2379	1806	89	3084	267
2013	2391	1983	189	3202	347
2014	2378	2167	141	3656	303
2015	1320	1032	124	1985	123
2016	3559	3015	283	4985	430
2017	2487	1867	191	3350	333
2018	2622	1768	166	3258	356
2019	2642	1764	128	3484	458
2020	2735	2174	198	4225	440
2021	2823	2167	153	3468	533
2022	2442	1914	155	3288	456

[43]: <Axes: title={'center': 'Share of Energy Discourse in Parliamentary Environmental Debates'}, xlabel='Year', ylabel='Proportion of Speeches'>



```
[52]: categories
      [extractivism]                23733
      [developmentalism]           16154
      [energy]                     11553
      [extractivism, developmentalism] 5383
      [extractivism, energy]         4260
      [extractivism, energy, developmentalism] 2535
      [energy, developmentalism]     2153
      [sustainability]              941
      [environmental_justice]        729
      [extractivism, energy, developmentalism, sustainability] 661
      [extractivism, developmentalism, sustainability] 620
      [developmentalism, sustainability] 586
      [extractivism, sustainability] 571
      [extractivism, environmental_justice] 418
      [extractivism, energy, sustainability] 287
      [energy, developmentalism, sustainability] 207
      [extractivism, energy, environmental_justice] 186
      [energy, sustainability]       165
      [extractivism, developmentalism, environmental_justice] 99
      [energy, environmental_justice] 95
      Name: count, dtype: int64
```

```
[105]: df[ENV_KEYWORDS.keys()].sum()
```

```
[105]: extractivism          40362  
       energy_projects      22299  
       developmentalist     27079  
       sustainability        33328  
       environmental_justice  1918  
       dtype: int64
```

```
[106]: env_cols = list(ENV_KEYWORDS.keys())  
  
df["env_speech"] = (df[env_cols].sum(axis=1) > 0).astype(int)
```

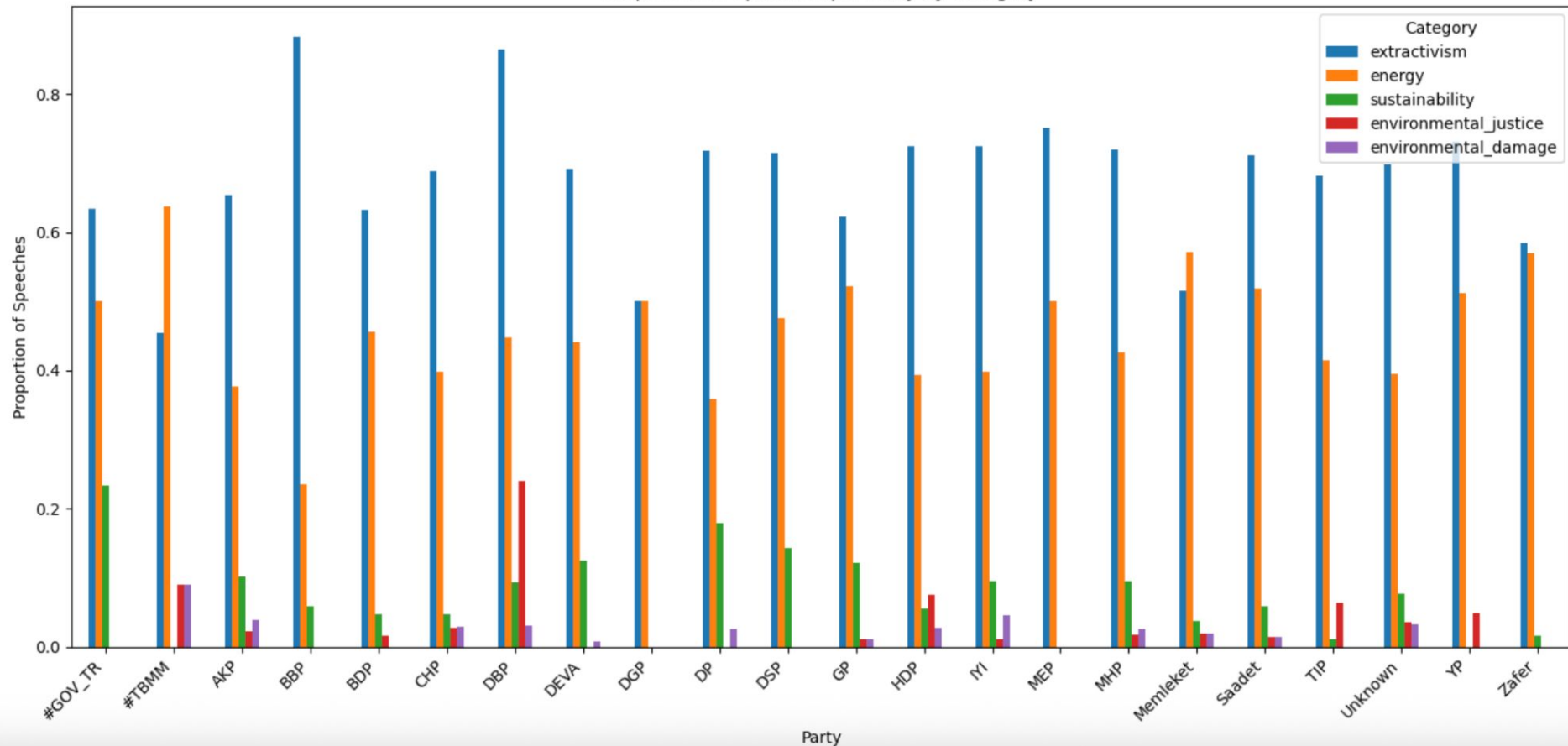
```
[107]: df["env_speech"].value_counts()
```

```
[107]: env_speech  
0      652365  
1       87379  
Name: count, dtype: int64
```

```
[61]: <Axes: title={'center': 'Gezi Park-Related Parliamentary Speeches (2011-2022)', xlabel='Year', ylabel='Number of Speeches'}>
```



Proportion of Speeches per Party by Category



Contribution

This research shows that parliament does more than responding to environmental conflict and struggles by analyzing the 1341 sessions in parliament. The parliament decide how that environmental struggle will be governed by majorly changing the discourse.